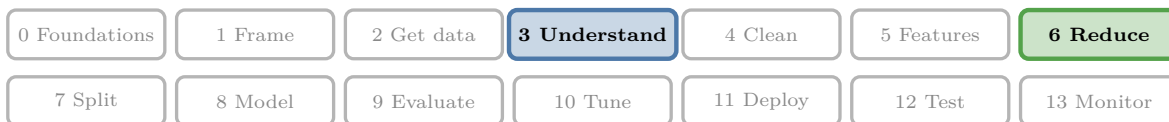


PCA: Geometric Intuition

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Where this fits: Pipeline map steps 3 (Understand data), 6 (Reduce dimensions). See the [Course map](#).



- **Builds on:** Dimensionality reduction ([Note ML-003](#)); Unsupervised learning ([Note ML-003](#)); Feature scaling ([Note ML-006](#)); Feature engineering ([Note ML-007](#)); Standardization ([Note ML-009](#)); Descriptive statistics ([Note ML-018](#)).
- **Leads to:** Covariance and covariance matrix ([Note ML-047](#)); One-way ANOVA ([Note MA-046](#)).
- **Compare with:** Feature construction and splitting ([Note ML-044](#)); Feature selection ([Note ML-045](#)); Low-rank approximation (truncated SVD) ([Note MA-059](#)).

1. What PCA is

Key point: PCA is an unsupervised feature extraction technique. PCA turns many features into a few new ones while keeping the essence of the data.

A **feature** (G-772) is an input variable (one column of the data table), an **observation** (G-1374) is one record (one row), and the **target** (G-1949) is the output we predict.

Principal component analysis (PCA) (G-1469) is the best-known feature extraction technique. PCA reduces the number of features in a dataset, which fights the curse of dimensionality ([Note ML-045](#)).

Three facts to keep in mind:

- PCA is **unsupervised** (G-2058): it uses only the features, never the target.
- PCA is old and widely used.
- The full mathematics of PCA is involved. This Note builds the geometric intuition; the step-by-step mathematics comes in the next Note.

1.1 The photographer

Key point: A photo turns a 3D scene into 2D. A good photographer picks the angle that keeps the most information, and PCA does the same with data.

A photographer at a football match has a 3D scene in front of them: players spread over a pitch. A newspaper photo is 2D, so every photo loses one dimension.

From some angles the photo is useless: the players stand in a line, one behind the other. From other angles every player is clearly visible. Figure 1 shows both cases.

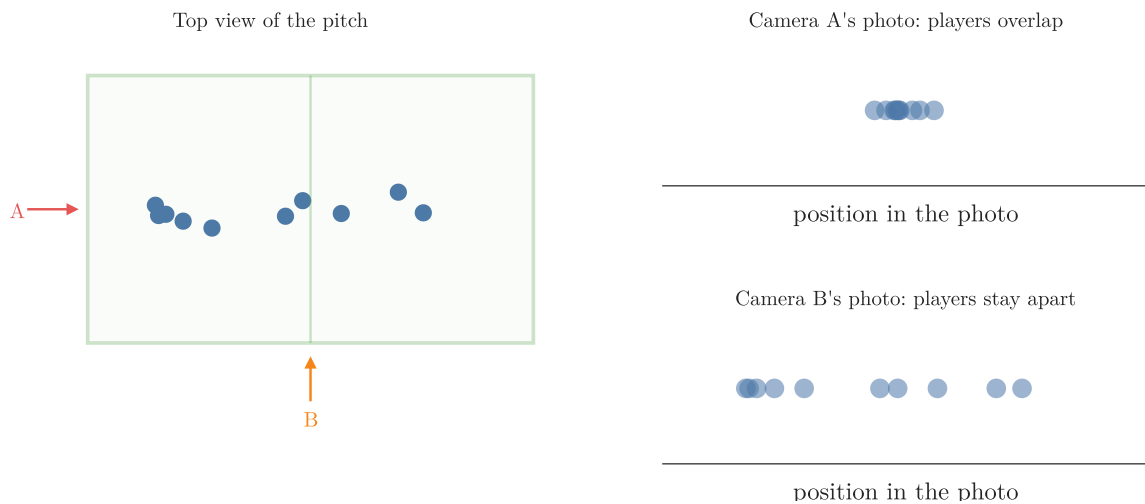


Figure 1: What two cameras see of the same 10 players

- **Camera A** looks along the length of the pitch. In its photo, the players overlap.
- **Camera B** looks across the pitch. In its photo, the players stay apart.

PCA works like a photographer who walks around the stadium to find the best angle. PCA moves data from a high dimension to a lower one, and picks the direction that keeps the most information.

1.2 Why use PCA

Key point: Fewer features make algorithms faster. Going down to 2 or 3 features lets us plot the data.

1. **Faster algorithms.** With fewer features, there is less data to process, both in training and in prediction. Performance can stay close to what it was on the full data; Note ML-048 measures this on images of digits.
2. **Visualisation.** We cannot plot more than 3 dimensions. PCA can bring a dataset with many features down to 3 or 2, and then we can plot it. A later Note does this with images of digits that have 784 features.

2. Feature selection by spread

Key point: To choose between two features, project the points onto each axis and keep the feature whose points are more spread out.

PCA is a feature extraction technique, but it is easiest to understand by first looking at feature selection: keeping some existing features and dropping the rest (see the [feature engineering Note](#), sections 8 and 9, for both parts).

Take made-up data about flats with three features (one observation per flat): the number of

rooms, the number of grocery shops nearby, and the price. Suppose we must drop one of the two input features.

Anyone who knows a little about property would keep rooms: the price depends much more on rooms than on nearby grocery shops. But often we work with data we know nothing about. We need a rule that does not depend on knowing the subject.

2.1 The rule: keep the bigger spread

Key point: The shadow of the points on an axis shows how much that feature varies. The longer the shadow, the more information.

Figure 2 (left) plots the flats with rooms on the x-axis and grocery shops on the y-axis.

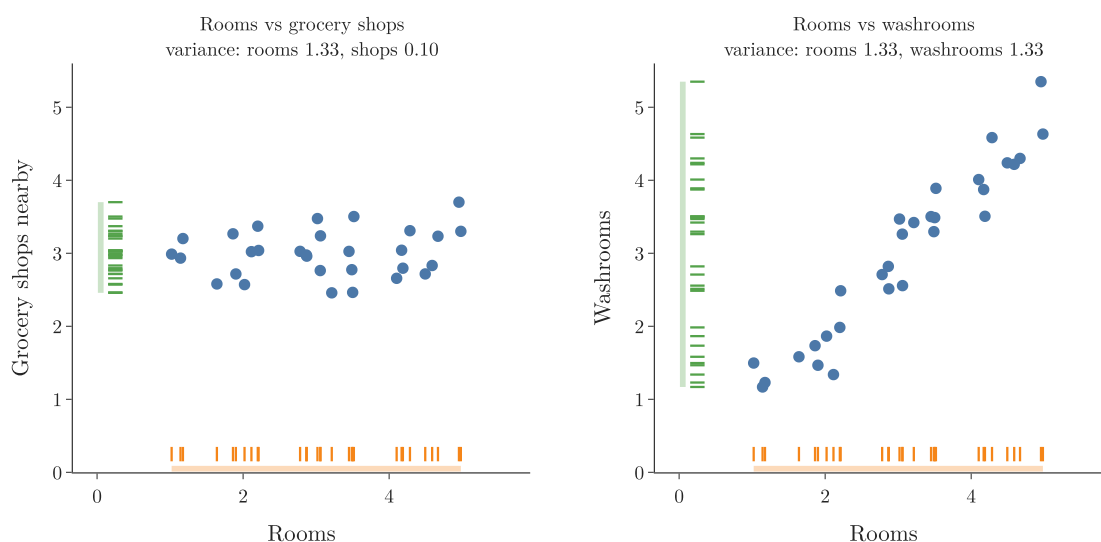


Figure 2: Shadows of the points on each axis

Imagine shining a light so each point casts a shadow on the x-axis. Dropping each point straight down onto an axis like this is called **projecting** (G-1583) it.

- On the rooms axis (orange), the shadows stretch from 1 to 5.
- On the grocery shops axis (green), they cover only a short stretch.

The data is spread out along rooms and bunched up along grocery shops. So we keep rooms. The size of this spread is measured by the **variance** (G-2074) of the shadows: 1.33 for rooms and 0.10 for grocery shops.

Feature selection by spread therefore keeps the features with the largest variance.

2.2 Where feature selection fails

Key point: When two features have about the same spread, feature selection cannot tell which one to drop.

Now replace grocery shops with the number of washrooms. Rooms and washrooms rise together: a flat with more rooms usually has more washrooms.

Figure 2 (right) shows this data. Projected onto either axis, the spread is the same: variance 1.33 for both. Both features matter for the price, and the rule from Section 2.1 cannot choose.

Feature selection is stuck. Feature extraction solves this.

3. Feature extraction: a new feature

Key point: Instead of dropping one of two equal features, build one new feature that carries the information of both.

The [types of ML Note](#) (section 3.3) already did this by hand: rooms and washrooms were replaced by one new feature, the flat's area. The data went from 2 input features to 1, and the price can still be predicted.

PCA does the same with no knowledge of the subject. PCA ignores the original features as they are, builds a new set of features from the data alone, and keeps the new features that matter most.

Figure 3 shows the result on the rooms and washrooms data. Every flat slides onto one line, PC1 (section 4 explains how PCA chooses it), and is then described by a single number: its position along that line, from -2.7 to $+3.0$. That one number keeps 98 percent of the spread of the two original features.

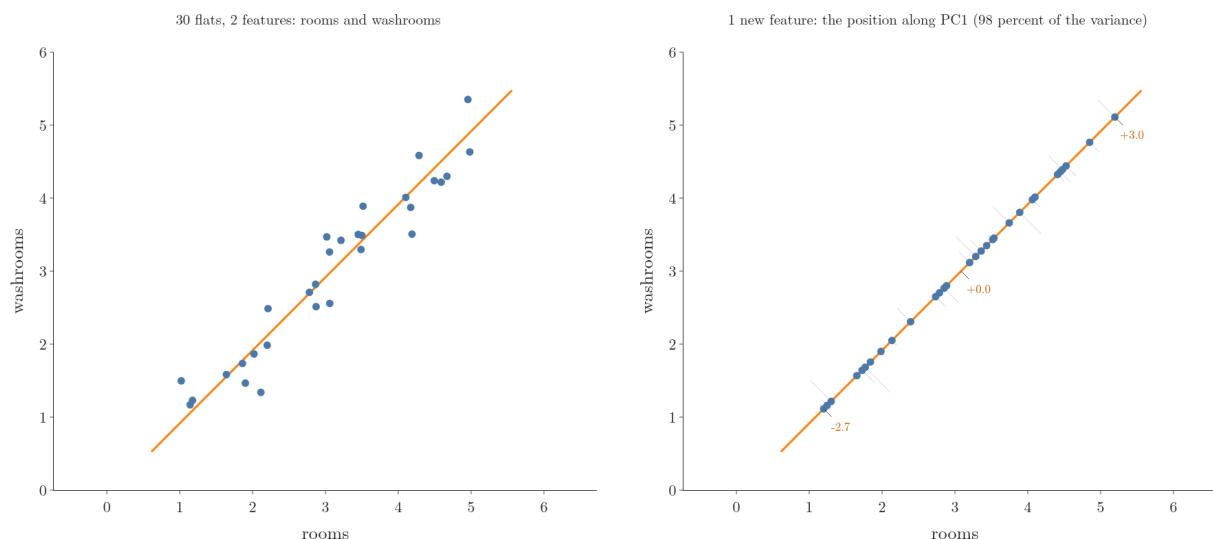


Figure 3: The 30 flats move onto the PC1 line (orange); each flat is then described by one number, its position along PC1, which keeps 98 percent of the variance.

4. How PCA finds the new features

Key point: PCA rotates the axes until one axis points along the direction of greatest spread. That axis is the first principal component.

Feature selection could only look along the two existing axes, rooms and washrooms. PCA is allowed to turn the axes to any angle.

Figure 4 shows the idea on the rooms and washrooms data. We draw a line through the centre of the data, project every point onto it, and measure the variance of the shadows. Then we turn the line and measure again.

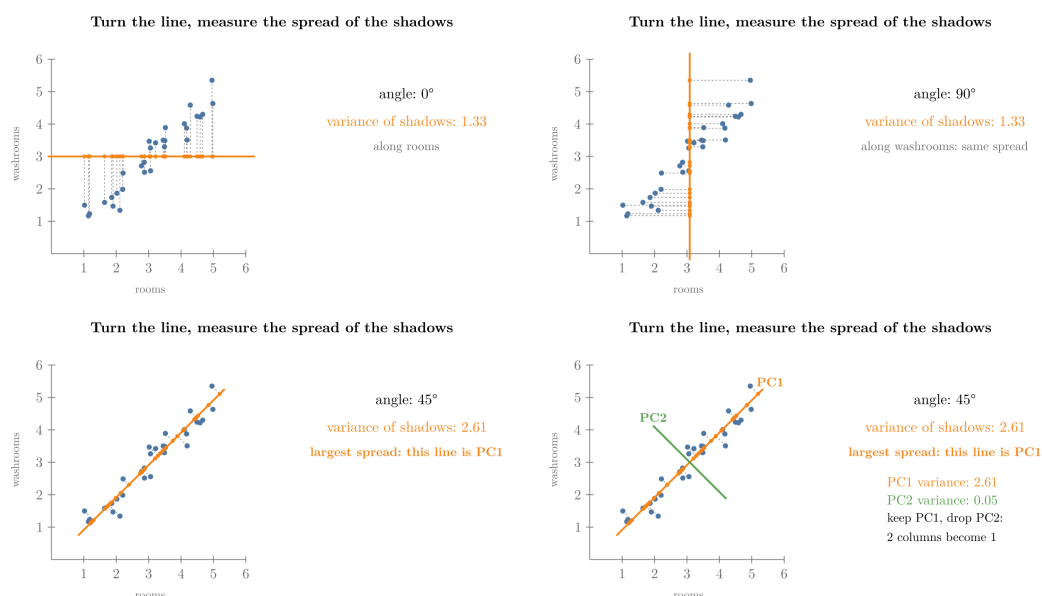


Figure 4: Turning the line until the shadows spread the most

1. **Along rooms (0°):** variance 1.33.
2. **Along washrooms (90°):** variance 1.33, the same.
3. **At 45°:** variance 2.61, the largest of any angle. This line is the first principal component.
4. **At right angles to it:** variance only 0.05. This line is the second principal component.

The new axes are called **principal components** (G-1563), written **PC1** and **PC2**. PC1 is the direction with the most variance; PC2 is at right angles to it and holds what is left.

Here PC1 holds almost all the spread, 2.61 against 0.05. So we keep PC1, drop PC2, and describe each flat by one number: its position along PC1. Two features have become one, just like the flat's area.

4.1 Largest spread and closest line are the same line

Key point: The line with the largest spread of shadows is also the line closest to the points. Pythagoras' theorem ties the two together.

“Find the line the shadows spread most along” and “find the line that passes closest to the points” sound like two different jobs. Figure 5 shows they are one job. It has two steps.

1. **Centre the data.** Subtract the mean of each feature, so the centre of the cloud sits on the origin. The data is now **centred** (G-365). The shape of the cloud does not change: the highest flat is still the highest.
2. **Turn a line through the origin** and project every flat onto it, as in Figure 4.

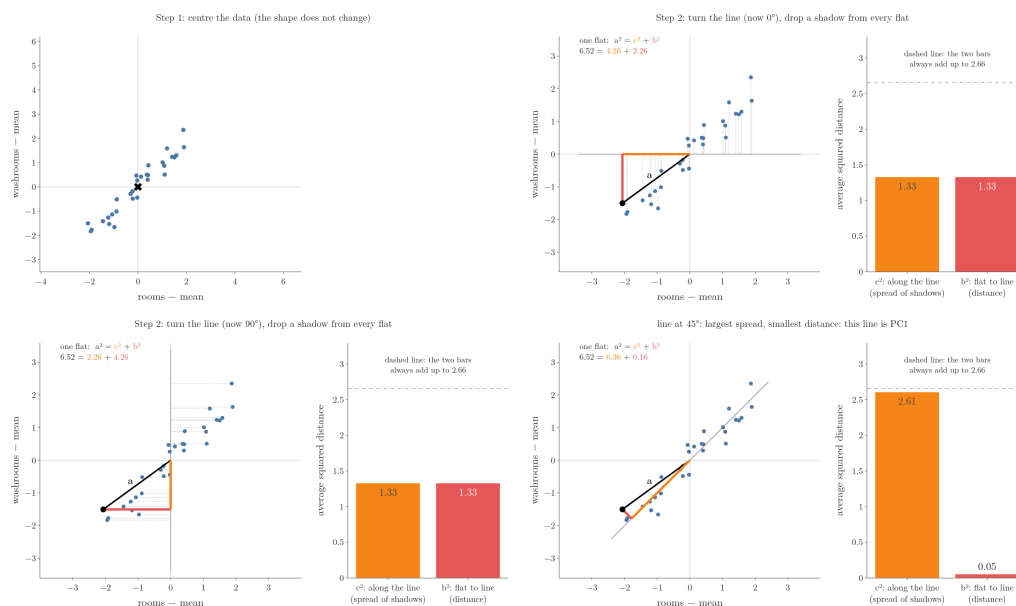


Figure 5: The centred flats and a line turning through the origin. For one flat (black), a is its distance from the origin, c (orange) the distance of its shadow from the origin, and b (red) its distance to the line. The bars average c^2 and b^2 over all 30 flats; they always add up to 2.66. Idea after StatQuest, “Principal Component Analysis (PCA), Step-by-Step”.

Watch the black flat in Figure 5. Three lengths form a right-angled triangle:

- a : from the origin to the flat. The flat does not move, so a never changes;
- c : from the origin to the flat's shadow on the line;
- b : from the flat to the line.

By Pythagoras' theorem,

$$a^2 = b^2 + c^2.$$

For the black flat, $a^2 = 6.52$ at every angle. At 0° it splits into $c^2 = 4.26$ and $b^2 = 2.26$. At

45° it splits into $c^2 = 6.36$ and $b^2 = 0.16$. Because a^2 is fixed, whenever c^2 grows, b^2 must shrink by the same amount.

The same holds for every flat, so it holds for the averages, which are the two bars in Figure 5. The orange bar is the variance of the shadows; the red bar is the average squared distance to the line. Together they always make 2.66. At 45° the orange bar is at its largest, 2.61, so the red bar is at its smallest, 0.05.

Maximum variance along the line and minimum distance to the line are therefore the same answer. PCA uses the first description because the variance of the shadows is easier to compute.

4.2 How many principal components

Key point: Data with n features has at most n principal components. We keep the first few.

Rotating the axes does not create extra axes. Data with 2 features gives 2 principal components; data with 10 features gives up to 10. The same idea works in 3, 4 or any number of dimensions.

The components come in order: PC1 holds the most variance, PC2 the next most, and so on. Reducing dimensions means keeping the first few and dropping the rest.

The order also tells us how to read a plot drawn on PC1 and PC2: a gap between two groups along PC1 matters more than a gap of the same size along PC2.

Python: PCA in scikit-learn (full code in the Notebook).

```
from sklearn.decomposition import PCA

pca = PCA(n_components=2).fit(P)      # P: rooms and washrooms
pca.components_[0]                   # PC1 direction: [0.707, 0.707],
    ↪ i.e. 45 degrees
pca.explained_variance_ratio_        # [0.981, 0.019]: PC1 holds 98% of
    ↪ the variance

# keep only PC1: two features become one
PCA(n_components=1).fit_transform(P).shape  # (30, 1)
```

How PCA finds PC1 without trying every angle is the subject of the next Note.

5. Variance

Key point: The mean says where the data is centred. Variance says how spread out it is.

PCA is built on variance, so it is worth being precise about what variance means.

5.1 Mean is not enough

Key point: Two datasets can have the same mean and very different spreads.

The mean gives the centre of the data (see the [understanding your data Note](#), section 7.1), but not its spread. Take two small datasets (Figure 6):

- Data A: $-5, 0, 5$
- Data B: $-10, 0, 10$

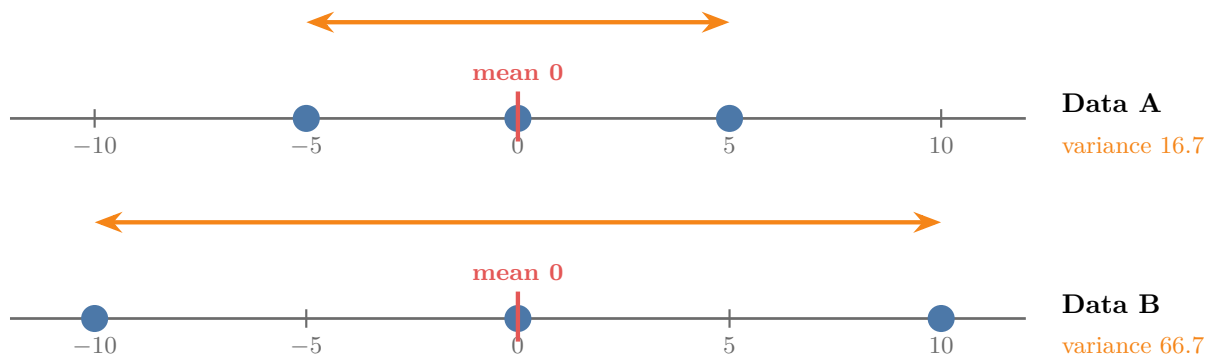


Figure 6: Same mean, different spread

Both have mean 0, yet Data B is clearly more spread out. To measure that we need variance.

5.2 The variance formula

Key point: Variance is the average squared distance of the points from their mean.

In words: for every point, find how far it is from the mean, square that distance, and average the squares.

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Here x_i is the i -th point, $\bar{x}$ is the mean and n is the number of points.

For Data A, with mean 0:

$$\sigma^2 = \frac{(-5)^2 + 0^2 + 5^2}{3} = \frac{50}{3} \approx 16.7$$

For Data B:

$$\sigma^2 = \frac{(-10)^2 + 0^2 + 10^2}{3} = \frac{200}{3} \approx 66.7$$

Data B's variance is 4 times Data A's. Variance tells the two datasets apart where the mean could not.

Extra: Here we divide by n , which gives the **population variance**. Dividing by $n - 1$ instead gives the **sample variance**, a slightly larger value. A sample tends to look a little less spread out than the population it came from; dividing by $n - 1$ corrects this exactly on average (Casella and Berger, Thm 5.2.6). NumPy divides by n by default (`ddof=0`); pandas divides by $n - 1$ (`ddof=1`), which is why the standard deviation in `describe` uses $n - 1$.

5.3 Variance and spread

Key point: Variance grows with spread but is not the spread itself. The square root of variance, the standard deviation, is in the same units as the data.

Data B is twice as spread out as Data A, but its variance is 4 times larger, because the distances are squared. The square root of the variance, the standard deviation (see the [understanding your data Note](#), section 7.1), is back in the units of the data: about 4.1 for Data A and 8.2 for Data B, exactly twice.

5.4 Why squares, not absolute distances

Key point: Squared distances give a smooth formula that the optimisation inside PCA can work with; absolute distances do not.

We could measure spread without squares: take the absolute distance $|x_i - \bar{x}|$ of each point from the mean and average them. That average is the **mean absolute deviation** (G-1193).

PCA does not use it. Finding the best direction is an optimisation problem, and solving it needs a formula we can differentiate. The absolute value has a sharp corner at zero, where it cannot be differentiated. The square is smooth everywhere, so variance is used (Figure 7).

Some measure with no sign is needed in the first place because the plain distances $x_i - \bar{x}$ are positive on one side of the mean and negative on the other, and they cancel: for Data A, $-5 + 0 + 5 = 0$. Squaring makes every distance positive, so nothing cancels.

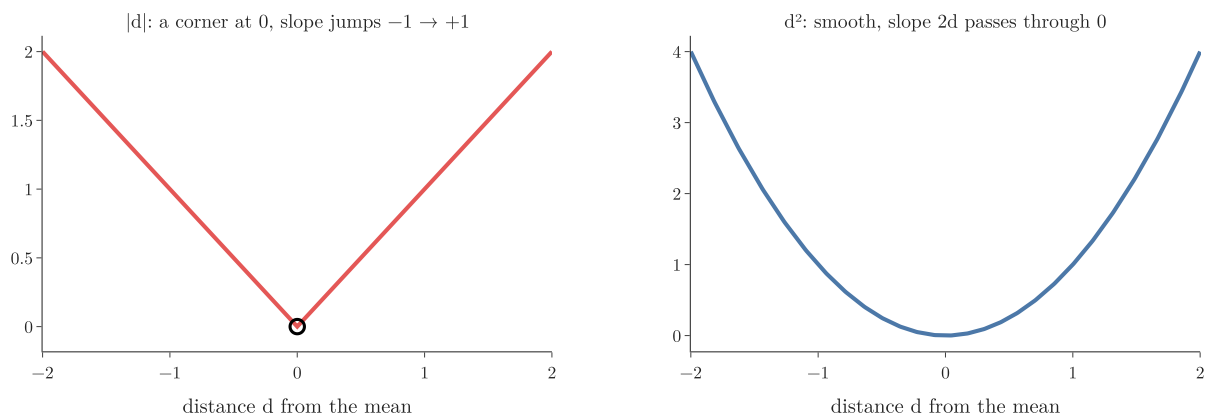


Figure 7: Left: the absolute distance $|d|$ has a corner at 0, where its slope jumps from -1 to $+1$. Right: the square d^2 is smooth, and its slope $2d$ passes smoothly through 0.

6. Why PCA maximises variance

Key point: Keeping the direction of greatest variance keeps the points as far apart as they really are. A low-variance direction squeezes different points together.

Figure 8 picks out two flats from the rooms and grocery shops data. They are 3.15 apart: one has 1 room and the other 4, with almost the same number of grocery shops.

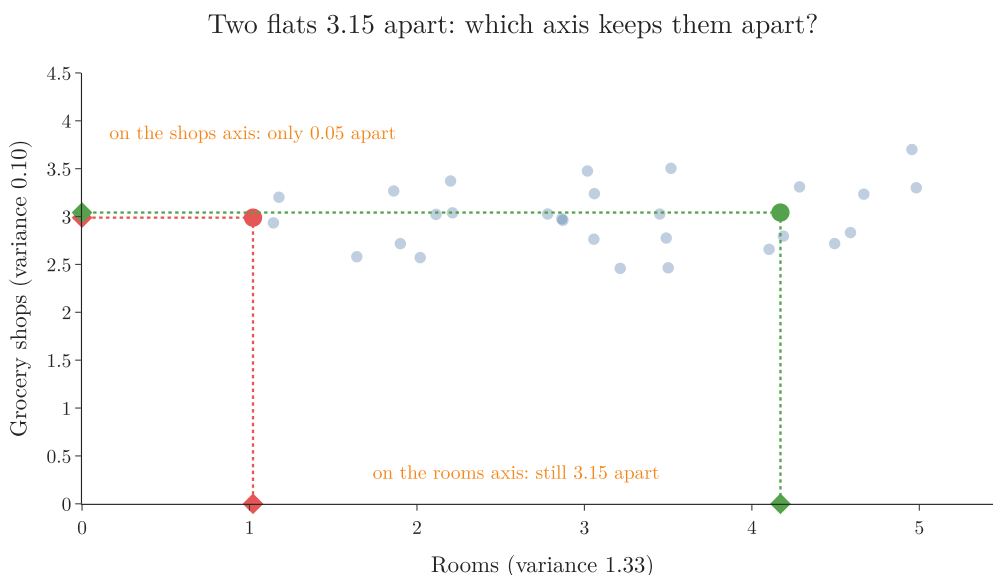


Figure 8: Two flats projected on each axis

- **Projected on rooms** (the high-variance axis), they are still 3.15 apart.
- **Projected on grocery shops** (the low-variance axis), they are only 0.05 apart. They look almost identical.

Many algorithms, such as KNN, work with distances between points. After a projection onto the grocery shops axis, such an algorithm could never tell these two flats apart.

PCA therefore looks for the direction of maximum variance. That direction keeps the distances between points, and so the relationships in the data, as close as possible to the original: the photographer choosing the angle where the players stay apart.

Section 4.1 showed the same choice from the other side: the direction of maximum variance is also the line closest to the points.

7. Summary

Idea	What it means
PCA	Unsupervised feature extraction: few new features from many old ones
Projection	Dropping each point onto an axis, like a shadow
Variance	How spread out the points are; average squared distance from the mean
Feature selection by spread	Keep the existing feature with the largest variance
Its weakness	Fails when features have similar variance (rooms and washrooms)
Principal components	New, rotated axes; PC1 has the most variance, PC2 the next
Why maximise variance	Points that are far apart stay far apart

- PCA makes algorithms faster and lets us plot high-dimensional data.
- Feature selection can only keep or drop existing features; PCA builds new ones.
- PCA rotates the axes so that PC1 points along the greatest spread.
- The line of greatest spread is also the line closest to the points (Pythagoras: $a^2 = b^2 + c^2$ with a fixed).
- Data with n features has at most n principal components; we keep the first few.
- Variance, not mean absolute deviation, because it is smooth enough to optimise.

8. Sources

Built from

- CampusX, “Principle Component Analysis (PCA) | Part 1 | Geometric Intuition”, YouTube, <https://www.youtube.com/watch?v=iRbsBi5W0-c>
- StatQuest with Josh Starmer, “StatQuest: Principal Component Analysis (PCA), Step-by-Step”, YouTube, <https://www.youtube.com/watch?v=FgakZw6K1QQ> (centring, the turning line and the Pythagoras triangle of section 4.1)
- StatQuest with Josh Starmer, “StatQuest: PCA main ideas in only 5 minutes!!!”, YouTube, https://www.youtube.com/watch?v=HMOI_lkzW08 (reading a PCA plot: PC1 gaps matter more than PC2 gaps)

Other references

- Casella, G. and Berger, R. L. (2002). *Statistical Inference*, 2nd edition, Theorem 5.2.6. Duxbury.

9. Key terms

Term	Meaning
Feature	An input variable: one column of the data table
Observation	One record: one row of the data table
Principal component analysis (PCA) (G-1469)	An unsupervised feature extraction technique that builds new features along the directions of greatest variance
Projection	Dropping each point onto an axis or line, like casting a shadow
Centred data	Data whose mean is 0: the mean of each feature has been subtracted
Variance (G-2074)	The average squared distance of the points from their mean
Mean absolute deviation	The average absolute distance of the points from their mean
Principal component	A new axis found by PCA; PC1 holds the most variance, PC2 the next most